



Figure 1: The snap execution environment

revisions of file systems under `/snap/foo` but `/snap/foo/current` always points to the desired version as per the recent upgrade or downgrade. Each user-accessible application of the snap is linked under `/snap/bin`, with the `<snap-name>.<app-name>` convention, e.g., `/snap/bin/foo.test` for an application named `test` under the snap `foo`. The meta data of the snap is available in the form of `meta/snap.yaml` under the mounted file system.

## Use cases

As the underlying stack is read only and each snap has a self-contained file system, Ubuntu Core is the perfect choice for industrial solutions for which security is a critical concern. Here are some cases of Ubuntu Core powered solutions:

- IoT gateways for industrial solutions, like the Dell Edge gateway
- DeviceHive IoT toolkit
- Cloud based servers like NextCloud
- Digital signage
- Robots and drones

## Some popular snaps

Here are some popular applications available from the snap store:

- Atom, a text editor
- NextCloud, a cloud based server for data storage
- VLC, a multi-media player
- Docker, a container manager
- Cassandra, a scalable NoSQL database
- Blender, a 3D creation suite
- Stellarium, a planetarium software
- Rocket.Chat, a Web chat server
- The Telegram desktop client
- Jenkins, a leading automation server for building,

testing and deployments

- OpenHAB, a home automation middleware software
- Mosquitto, a MQTT broker
- Clients for AWS, like Azure cloud solutions

## Installation

Ubuntu Core comes with a snap manager, `snappy`, which is present by default in Ubuntu Xenial (16.04 LTS) onwards; for older versions of Ubuntu and other distributions you can install the snap manager as follows:

```
apt-get install snappy
```

If you are planning to build custom snaps, install `snappy` also, as follows:

```
apt-get install snappy
```

Replace `apt-get` with any other package manager like `yum`, `dnf`, `zypper`, `pacman`, `poky`, etc, for non-Debian based distributions.

Ubuntu Core is available for dedicated installation for various embedded targets like Raspberry Pi Models 2 and 3, Compute Module 3, Intel Joule, Dragon Board, Intel NUC kit, Samsung Artik, etc. Also, Ubuntu Core can be installed through KVM on any desktop as a virtual machine, for which you can download the core image for the AMD 64 architecture, extract the archive and run the following command:

```
qemu-system-x86_64 -snap 2 -m 1500 -netdev user, id=mynet0, hostfwd=tcp::8022::22, hostfwd=tcp::8090::80 -device virtio-net-pci, netdev=mynet0 -drive file=ubuntu-core-16-amd64.img, format=raw
```

Now log in to the core using ssh as follows:

```
ssh username@localhost -p 8022
```

## Working with snaps

Let's try working with snaps in CLI mode, assuming `osfy` as the snap's name.

To search for available snaps based on the `osfy` keyword, use the following command:

```
snap find osfy
```

To install a specific snap from the store, use the command given below:

```
snap install osfy
```

Observe the file hierarchy in `/snap/osfy/current`. Run the application associated with snap by using